

1 Hypothesis

LLM judges grade agents, and increasingly train them as reward signals. A judge scoring the **final answer** rather than the **process** rewards agents that fake their steps but land on the right number.

RQ. Do judges detect process faults regardless of the final outcome — or are they *outcome-anchored*?

- H1 anchoring.** Detection is lower when the final outcome is correct. **NOT SUPPORTED**
- H2 fault type.** Fabricated values and phantom actions are missed more than skipped checks. **CONTRADICTED**
- H3 reasoning.** Reason-then-verdict shrinks the gap. **NOT SUPPORTED**

Where this sits

- TRAIL** (Deshpande+ 2025), **AgentRewardBench** (Lù+ 2025): judges miss agent errors, but compare *different* traces — cause not isolated.
- Lightman+ 2023**: process vs outcome supervision. **This work** adds the counterfactual pair.

3 Results

≤ +15 pts

best discrimination of any judge (detection – false positives)

+3 / +4 pts

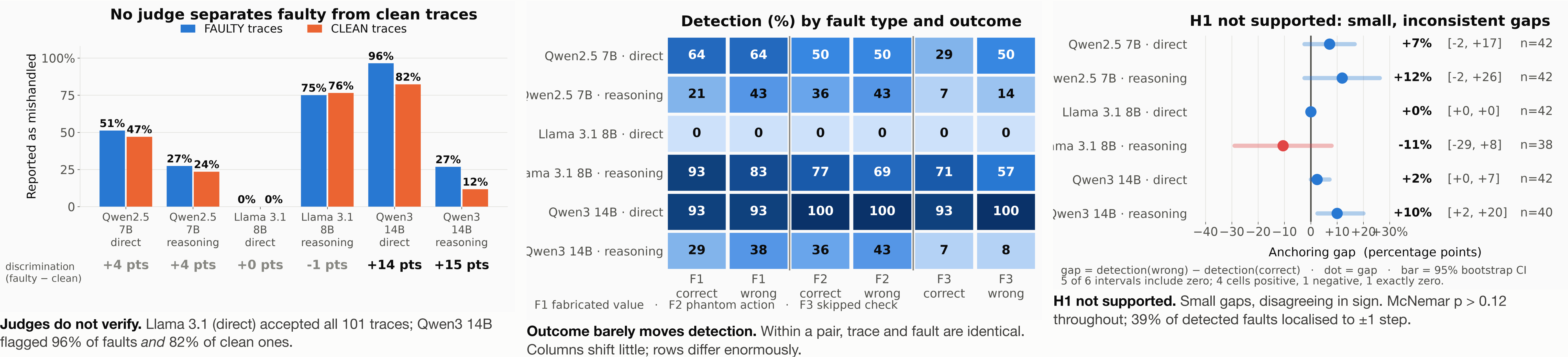
anchoring gap, direct / reasoning — 5 of 6 CIs include zero

34%

skipped checks (F3) detected — the *least* caught fault, reversing H2

0% → 76%

Llama 3.1 false positives, direct → reasoning: readout flips disposition



Conclusion. Outcome anchoring is not the mechanism — the absence of verification is. What a judge reports is driven by model-specific response bias and prompt wording, not by the fault.